General Solutions

Exam code: 9MA0

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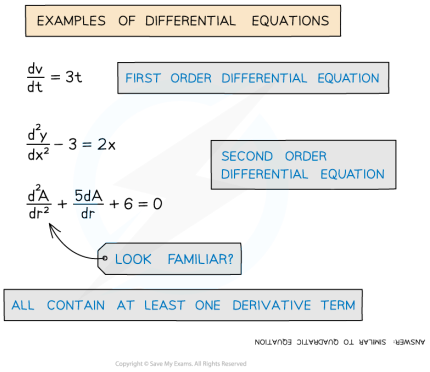
General solutions

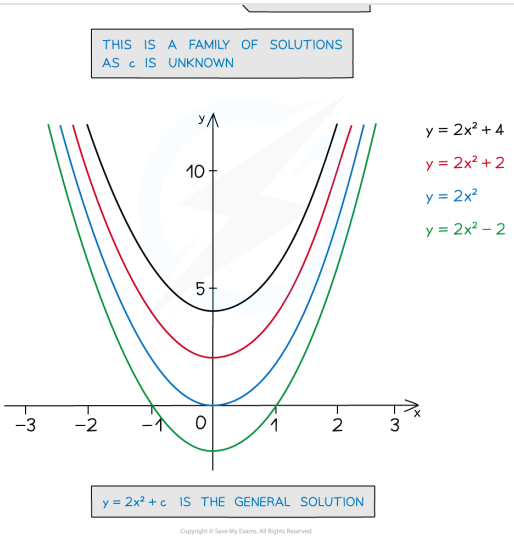
What is a differential equation?



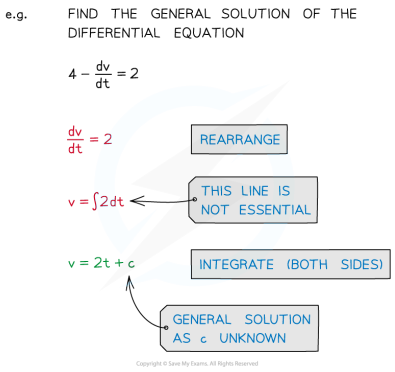
- Any equation, involving a **derivative term**, is a **differential equation**
- Equations involving only first derivative terms are called **first order differential equations**
- Equations involving second derivative terms are called **second order differential equations**

What is a general solution?

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

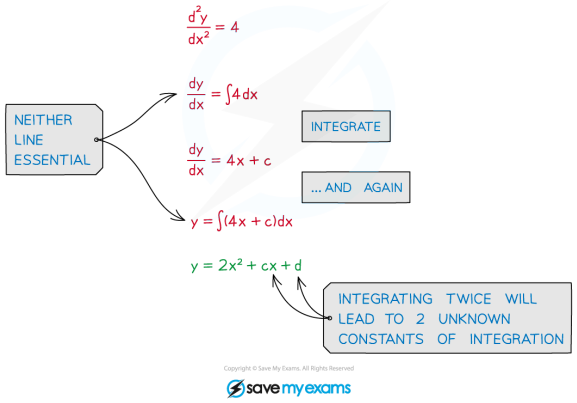


- Integration will be involved in **solving** the differential equation
  - ie working back to "**y = f(x)**"
- A constant of integration, **c** is produced
- This gives an infinite number of solutions to the differential equation, each of the form **y = g(x) + c** (ie **y = f(x)** where **f(x) = g(x) + c**)
- These are often called a **family of solutions** ...
  - ... and the solution **y= g(x) + c** is called the **general solution**



Worked Example

equation $\frac{d^2y}{dx^2} = 4$



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